

General Education Reflection

Throughout my four years at Iowa State University, I found out that a successful electrical engineer is not limited to their engineering expertise. The classes in my engineering curriculum taught me how to analyze circuits and technical problems. Classes outside my engineering major allowed me to see the problem from other perspectives. They taught me that engineering decisions have impacts on people, societies, companies, and the environment. Thinking back, ANTHR 230, LD ST 322, ECON 101, and SP CM 212 were very useful in shaping my skills as an electrical engineer in the future.

One course that changed my perspective is ANTHR 230 (Globalization and Human Condition). This course made me realize that people live under different conditions in the world. One lesson that I learned in this course is that a solution that works well in one situation does not necessarily work well in others. For instance, in electrical power systems, engineers not only try to provide reliable electricity, but they also need to make sure that the technology is affordable, accessible, and practical for the communities that will use it. This is extremely important for developing countries where some communities do not have access to electricity. Since I want to pursue a career in power systems, I learned through this course that engineering is not only about making things work but also about making sure that things will help the people in need.

Besides that class, another class that has been useful to me is LD ST 322 (Leadership Styles and Strategies in a Diverse Society). It helped me realize that there are lots of diverse opinions, means of communication, and problem-solving techniques. An important characteristic of a good leader is not to do everything on their own, but to try to understand other people's opinions and unite the team around one goal. As for my experience, I can relate LD ST 322 to my Senior Design project. My team is designing an eddy current probe in order to test metal electrical conductivity. People in a team are different, and that is why it is necessary to share ideas and distribute tasks among team members.

Learning ECON 101 (Principles of Microeconomics) gave me a different view on engineering problems after examining a given solution only in terms of effectiveness and performance in the past. However, in this class, I understood that cost, resource consumption, and economic considerations play the same role in decision-making. For instance, in the power industry, engineers have multiple choices of technologies that could be applied to design and upgrade an electrical system. Yet the most technologically sophisticated technology is not necessarily the optimal one due to its high cost of

acquisition and maintenance. Engineers should be able to make a compromise between performance, reliability, and costs. This course confirmed what I learned in my power systems courses at Iowa State. Economic aspects are very important when it comes to designing and operating electrical power systems.

The SP CM 212 course (the fundamentals of Public Speaking) demonstrated to me how important communication skills are in the world of engineering. Although an engineer could come up with a wonderful idea, failing to explain those ideas could mean failure for the whole process. By taking this course, I learned how crucial it was to express oneself well, articulate clearly, and think about the audience. I saw the importance of communication skills during my internship at Electrical Power Products as a Quality Engineering Intern. My main responsibility was to work on engineering instructions so that workers could understand how to work with electrical appliances. Thanks to that experience, I understood that communication is necessary not only for making presentations but for being successful in teamwork and technical writing too.

When combining these three courses, the first thing that I see is that engineering problems can be more than technical. First, as the whole world strives towards renewables, engineers will have to consider not only efficient generation of electricity. They should pay attention to the expenses of the necessary infrastructure, environmental impact, and how these changes may affect various groups of people. Another aspect is providing reliable electricity access to all people in the world. There are solutions to this problem with engineering, but still engineers will have to consider financial restrictions, resources available, and also future users of the product.

Courses on general education have given me irreplaceable knowledge that I can use for my career. They have widened my understanding beyond calculations to the people who affect engineering decisions. This will be particularly useful as I transition from Iowa State University into my career in electrical engineering. In any work I do in this field, from dealing with power systems to collaborating with other engineers to talking with clients. The knowledge gained in these courses will make me a more responsible, open-minded, and skilled professional.